C Programming – 01

Assignment 01: Increment and Decrement Operators

1. What will be the output of the following C program?

```
#include <stdio.h>
int main()
{
    int a = 10, b = 5;
    printf("%d %d\n", a++, --a);
    printf("%d %d\n", ++b, b++);
    return 0;
}
```

2. Find the final values of a, b, and c.

```
int x = 10, y = 5;
int a = x++;
int b = y++;
int c = ++x + --y;
```

3. What will be the output?

```
int a = 5, b = 10;
a++;
b--;
++a;
--b;
printf("%d %d", a, b);
```

3. Explain the difference between pre-increment (++a) and post-increment (a++) with suitable examples.

4:

1. Write a C program to check whether a given number is a Strong Number.

2. Write a C program to print all Strong Numbers between 1 and 10,000.

5:

1. Write a C program to check whether a given number is an Armstrong Number.

2. Write a C program to print all Armstrong Numbers between 1 and 10,000.

6:

1. Write a C program to check whether a given number is a Perfect Number.

7: Write a C program to find both the GCD and LCM of two integers using the relationship:

LCM(a, b) = (a × b) / GCD(a, b)

8. Write a C program to find the GCD of two numbers using the Euclidean algorithm.

9: Prime Number

1. Write a C program to check whether a given integer is a Prime Number.

2. Write a C program to print all Prime Numbers between 1 and 100.

10. Output problem:

```
#include<bits/stdc++.h>
using namespace std;

void fun(int i)
{
   if(i<=0)return;
   cout<<i--<<' ';
   fun(--i);
   cout<<i<<' ';
```

```
}
int main()
{
    fun(5);
}
```